MILITARY SPECIFICATION

YOKE ASSEMBLY: 10540507

1. SCOPE

1.1 This specification covers the assembly which supports the transceiver and provides elevation control and electrical contact for the Range Finder, Laser, AN/GVS-3 (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on the date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts, Equipment, and Tools for Army Materiel, Packaging and Packing of
MIL-I-45607	Inspection Equipment, Acquisition, Maintenance and Disposition of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-810	Environmental Test Methods

DRAWINGS

U.S.Army, Frankford Arsenal

F10540507	Yoke Assembly
F10540511	Housing
C10540587	Connector
F10544048	Gage, Functional (Telescope Assembly)
F10540942	Body
F10552431	Gage, Interchangeability, Maximum
F10552432	Gage, Interchangeability, Minimum

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PACKAGING DATA SHEETS

10540507 Packaging of Yoke Assembly

(Copies of specifications, standards, drawings, and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials.- Materials shall be in accordance with drawings, material specifications and general specifications forming a part of this specification.

3.2 Fabrication.- The yoke assembly shall be manufactured in accordance with Drawing F10540507 and drawings pertaining thereto.

3.3 General specification.- The yoke assembly shall meet the following requirements of MIL-F-13926:

(a) Dimensions and tolerances.
(b) Inorganic protective surface finishes.
(c) Electrical and electronic assemblies.
(d) Soldering electrical connections.
(e) Part identification and marking.
(f) Workmanship.

3.4 Environmental.

3.4.1 Storage temperatures.- The yoke assembly shall withstand thermal stabilization at temperatures of -80°Fahrenheit (F) and +160°F. Subsequent to exposure to the temperatures specified herein, the yoke assembly shall show no evidence of physical damage, and shall meet the requirements of 3.4.2 and 3.6 through 3.7.

3.4.2 Operational temperatures.- The yoke assembly shall meet the requirements of 3.5.1 through 3.5.7 when operated at a thermally stabilized condition at temperatures of -65°F, +155°F and at a room temperature of between +70° and +80°F.

3.4.3 Shock.- The yoke assembly shall withstand a total of 18 shock impulses, 3 in each direction of 3 mutually perpendicular directions. Each shock impulse shall be a half-sine wave with a time duration of 11 ± 1 milli-seconds. The peak amplitude of each shock impulse shall be 75 g's. Subsequent to shock, the yoke assembly shall show no evidence of physical damage and shall meet the requirements of 3.4.2 and 3.6 through 3.7.

3.4.4 Vibration.- The yoke assembly shall withstand vibration for 15 minutes each in 3 mutually perpendicular directions. Vibration in each direction shall be at frequencies varied logarithmically from 5 to 500 cycles.

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per second (cps). The double amplitude of applied vibration shall be 0.036 inch at lower frequencies and shall be maintained at, but not exceed, a 5g level at higher frequency. Subsequent to vibration, the yoke assembly shall show no evidence of physical damage, and shall meet the requirements of 3.4.2 and 3.6 through 3.7.

3.5 Performance.- The yoke assembly shall meet the requirements of 3.5.1 to 3.5.7 when used in conjunction with the following representative components of the range finder:

(a) Adapter mount (Gage 10552431 and 10552432)
(b) Telescope assembly (Gage 10544048)

3.5.1 Locking torque (yoke assembly to adapter mount).- The torque required to rotate the locking knob of an adapter mount through maximum clockwise excursion, to secure the yoke assembly to the adapter mount, shall be 10 ± 1 inch pounds.

3.5.2 Unlocking torque (yoke assembly from adapter mount).- The torque required to rotate the locking knob of an adapter mount, through the maximum counter clockwise excursion, to release the yoke assembly from the adapter mount shall not exceed 12 inch-pounds.

3.5.3 Elevation torque (yoke assembly to telescope assembly).-With a telescope assembly mounted in the yoke assembly (the top caps of the yoke assembly secured in position), the torque required to move the telescope through its elevation travel shall not exceed 1 inch-pound applied to the elevation knob of the yoke assembly at temperatures between 70° and 80°F. The torque shall not exceed 1.5 inch-pounds at the temperature extremes of -65° and +155°F.

3.5.4 Elevation backlash (yoke assembly to telescope assembly.- With a telescope assembly secured in the yoke assembly, backlash between the worm of the yoke assembly and the worm wheel of the telescope assembly shall not exceed 0.8 mils as read on the knob scale of the yoke assembly.

3.5.5 Lateral displacement (yoke assembly to telescope assembly).- With a telescope assembly mounted in the yoke assembly, and with the caps of the yoke assembly secured in the locked position, lateral displacement of the telescope assembly shall not exceed 0.10 mil, when the elevation knob of the yoke assembly is rotated through at least two revolutions clockwise and two revolutions counterclockwise.

3.5.6 Instrument light.- With the circuit for the instrument light energized by 28 volts, actuation of the light switch shall provide illumination to the elevation knob dial and to the plastic rods for illumination of the azimuth scale and level vial of the adapter mount.

3.5.7 Cable assembly.- The cable assembly of the yoke assembly shall withstand bending through an angle of at least 270 degrees around a 1/2 inch diameter mandrel. There shall be no evidence of physical damage or electrical failure.

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3.6 Electrical.- The electrical wiring of the yoke assembly shall conform to the schematic Drawing F10540515.

3.6.1 Continuity.- Each conductor of the cable shall be electrically continuous (nominally 0 resistance) between the appropriate terminations as specified on the electrical schematic drawing.

3.6.2 Insulation resistance.- The insulation resistance from each pin to every other pin and to the connector body shall not be less than 1000 megohms when subjected to a minimum potential of 500 volts, direct-current.

3.7 Interface requirements.- The yoke assembly shall interface with:

(a) Journal bearings of Housing 10540511 on telescope assembly to achieve requirements specified under torque (3.5.3), elevation backlash (3.5.4) and lateral displacement (3.5.5).

(b) Connector 10540587 (13J6) of telescope assembly to provide electrical contact.

(c) Mounting surface, key and mounting holes on Body 10540942 of microscope assembly.

(d) Mounting surface and locating pins of adapter mount. The yoke assembly shall be capable of being securely locked in position and provide continuity through the electrical contacts between the yoke assembly and the adapter mount when the locking knob of the adapter mount is rotated with a torque specified in 3.5.1 to its maximum clockwise excursion. When the locking knob is rotated to its maximum counterclockwise excursion, the yoke assembly shall be capable of being removed without restraint from the adapter mount.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions.- The component and subassembly inspection requirements of MIL-F-13926 form a part of the quality assurance provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 First article (initial production) approval.- The requirement for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of three (3) assemblies

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manufactured in the same manner using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4.2.1 Government testing.- When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to insure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for materials.

4.2.2 Contractor testing.- When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certificates of conformance for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection.

4.3 Inspection provisions.

4.3.1 Submission of product.- Unless otherwise specified by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105.

4.3.2 Examination and tests.

4.3.2.1 Components and subassemblies.- All components and sub-assemblies shall be inspected in accordance with the inspection provisions contained in Supplementary Quality Assurance Provisions (SQAP) listed in the technical data package (TDP). In the absence of SQAP's, the applicable Quality Assurance Provisions of MIL-F-13926 shall apply. Examination and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105 and the sampling plans specified in Tables I, II and III herein. Examination and tests for packaging, packing, and marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects in Tables I, II and III shall constitute the minimum inspection to be performed by the supplier after first article approval and prior to Government acceptance or rejection by item or lot.

TABLE I - CLASSIFICATION OF DEFECTS
(100% INSPECTION)

		REQUIREMENT	TEST PROCEDURE
101.	Locking torque	3.5.1	4.6.2.1
102.	Unlocking torque	3.5.2	4.6.2.2
103.	Elevation torque	3.5.3	4.6.2.3

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TABLE I - CLASSIFICATION OF DEFECTS, CONT'D:
(100% INSPECTION

		REQUIREMENT	TEST PROCEDURE
104.	Elevation backlash	3.5.4	4.6.2.4
105.	Lateral displacement	3.5.5	4.6.2.5
106.	Instrument light	3.5.6	4.6.2.6
107.	Cable assembly	3.5.7	4.6.2.7
108.	Continuity	3.6.1	4.6.3.1
109.	Insulation resistance	3.6.2	4.6.3.2
110.	Interface requirements	3.7	4.6.4

MINOR: NONE

NOTE: The tests in Table I shall be conducted at a temperature between +70°F and +80°F.

4.3.3 Acceptance and rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.4 Special sampling.

4.4.1 General.- Three yoke assemblies shall be selected at random as a special sampling from each 100 produced. The sample shall meet the requirements and tests in Table II.

TABLE II

		REQUIREMENT	TEST PROCEDURE
301.	Materials	3.1	Applicable dwgs & specs - visual
302.	Fabrication	3.2	Applicable dwgs & specs - visual
303.	General specification	3.3	MIL-F-13926 - visual

NOTE: The tests in Table II shall be conducted at a temperature between +70°F and +80°F.

4.4.2 Environmental.- Three yoke assemblies shall be selected at random as samples from each 50 produced, or from each month's production, whichever occurs first. Each sample shall be subjected to the tests in Table III and shall then meet the requirements and tests in Table I.

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TABLE III

		REQUIREMENT	TEST PROCEDURE
304.	Storage temperatures (-80°F and +160°F)	3.4.1	4.6.1.1
305.	Operational temperatures (-65°F and +155°F)	3.4.2	4.6.1.2
306.	Shock (between +70° and +80°F)	3.4.3	4.6.1.3
307.	Vibration (between +70° and +80°F)	3.4.4	4.6.1.4

4.4.3 Failure of sample.- Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the represented inspection lot shall be suspended by the Government until necessary corrections have been made by the contractor and the resubmitted items have been approved (see 4.3.3).

4.5 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government furnished inspection equipment.- Where the contract provides for Government furnished test equipment, supply and maintenance of test equipment shall be in accordance with applicable requirements specified in MIL-I-45607.

4.5.2 Contractor furnished inspection equipment.

4.5.2.1 Government design.- All inspection equipment specified by drawing number in specifications or SQAP forming a part of the contract shall be supplied by the contractor in accordance with technical data included in the technical data package list.

4.5.2.2 Contractor design.- The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.6 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.5.2.3 Inspection equipment design requirements.

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4.5.2.3.1 Adapter mount.- The special testing equipment utilized for positioning and securing the yoke assembly shall conform to Drawings F10552431 and F10552432.

4.5.2.3.2 Telescope assembly.- The special testing equipment utilized for testing the yoke assembly to telescope assembly shall conform to Drawing F10544048.

4.6. Test methods and procedures.

4.6.1 Environmental.- The assembly shall be thoroughly examined for physical defects after each environment test. A visual and tactile examination shall be made for missing parts and physical damage. Upon completion of all environmental tests, the assembly shall be subjected to the tests specified in 4.6.2 through 4.6.4.

4.6.1.1 Storage temperatures.- The test equipment utilized in this test shall be in accordance with the "Test Facilities" requirements of MIL-F-13926. The test shall be performed as specified in Procedure II of MIL-F-13926 at the temperature specified in 3.4.1. The yoke assembly shall be exposed to each specified temperature extreme for a minimum of 5 hours.

4.6.1.2 Operational temperatures.- The test shall be performed in accordance with Procedure I of MIL-F-13926 and conditions of 3.4.2 utilizing the testing equipment indicated in 4.6.1.1. The yoke assembly shall be exposed to each specified temperature for a minimum period of 5 hours. At the end of each 5 hour period, and while still at the specified temperature, the yoke assembly shall be subjected to the tests in 4.6.2 through 4.6.2.7. The yoke assembly shall conform to the requirements of 3.4.2.

4.6.1.3 Shock.- The test equipment utilized in this test shall be in accordance with the "Test Facilities" requirements of MIL-F-13926. The test shall be performed as specified in Procedure II of MIL-F-13926 except the shock pulse shall comply with the wave form, time duration and peak "G" value specified in 3.4.3.

4.6.1.4 Vibration.- The test equipment utilized in this test shall conform to the vibration apparatus requirements specified in MIL-STD-810. The yoke assembly shall be firmly mounted to the vibration table. The vibration equipment shall be capable of providing the conditions of acceleration amplitude, frequency and duration specified in 3.4.4. The vibration test shall then be performed in accordance with Method 514, Procedure I, Part I of MIL-STD-810.

4.6.2 Performance of the yoke assembly.- The following tests shall be performed with the yoke assembly positioned on special testing equipment that conforms to the requirements of 4.5.2.3.1. The special testing equipment that conforms to 4.5.2.3.2 shall be positioned in the yoke assembly.

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4.6.2.1 Locking torque (yoke assembly to adapter mount).- Standard torque measuring equipment with an appropriate adapter to accommodate the locking knob of the adapter mount gage shall be utilized to measure the required torque. With the adapter affixed to the locking knob, rotate the knob through the excursion specified in 3.5.1. The running torque required to rotate the knob shall not exceed the values specified in 3.5.1.

4.6.2.2 Unlocking torque (yoke assembly from adapter mount).- Conduct this test in sequence to the test in 4.6.2.1, utilizing the test equipment therein. Rotate the locking knob through the excursion specified in 3.5.2. The running torque required to rotate the knob shall not exceed the values specified in 3.5.2.

4.6.2.3 Elevation torque.- Standard torque measuring equipment with an appropriate adapter to accommodate the elevation knob of the yoke assembly shall be utilized to measure the required torque to rotate the telescope assembly gage. With the adapter affixed to the knob, the running torque required to rotate the telescope assembly gage shall not exceed the values specified in 3.5.3.

4.6.2.4 Elevation backlash (yoke assembly to telescope assembly.- Perform this test utilizing the telescope assembly gage secured in the yoke assembly. A wall target containing a plumb line and a horizontal zero line shall be utilized with this test. The vertical reticle line of the telescope gage shall be plumb. Position the elevation knob of the yoke assembly to its midpoint of travel with backlash excluded for the direction of travel. Observe through the viewing telescope gage and establish coincidence of the intersecting point of the vertical and horizontal reticle lines of the telescope gage to the intersecting point of the vertical and horizontal target image lines. Record the position of the elevation knob scale. Rotate the knob to its clockwise stop, then with backlash included, re-establish coincidence of the telescope gage reticle with respect to the target image without overtravel. Record the position of the elevation knob scale. Any difference between the two recorded measurements is the amount of backlash. Backlash shall not exceed the tolerance specified in 3.5.4, as measured on the elevation knob scale.

4.6.2.4.1 Elevation backlash-operating temperature.- The following test shall be performed with the yoke assembly positioned in a temperature test chamber that conforms to the "Test Facilities" requirements of MIL-F-13926 and conditions of 3.4.2. The test chamber shall be provided with an optically corrected viewing window and exit window to permit viewing through the telescope assembly. The test shall be performed utilizing the test procedures specified in 4.6.2.4, except the test chamber shall be designed to permit rotation of the elevation knob and viewing of the elevation knob scale to determine conformance with 3.5.4. During this test at low and high temperature, the viewing and exit windows of the test chamber shall be frost and moisture free respectively so as not to influence any mechanical measurement required for backlash as specified in 3.5.4 while viewing through the telescope assembly.

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NOTE: Should overtravel occur during the rotation of the elevation knob, the test is nullified. The test for backlash shall then be repeated.

4.6.2.5 Lateral displacement (yoke assembly to telescope assembly).- Conduct this test from the recorded position established in 4.6.2.4. Rotate the elevation knob at least two revolutions clockwise and two revolutions counterclockwise. Lateral displacement between the vertical reticle line of the telescope gage with respect to the vertical target image line shall not exceed the tolerance specified in 3.5.5, as measured on the target.

4.6.2.5.1 Lateral displacement - operating temperatures.- Perform the following test utilizing the testing equipment specified in 4.6.2.4.1 and conditions of 3.4.2. The test shall be performed as specified in 4.6.2.5 to determine conformance with 3.5.5.

4.6.2.6 Instrument light.- Energize the circuit of the instrument light with the voltage specified in 3.5.6. Actuation of the light switch shall provide illumination to the elevation knob dial and to the plastic rods for illumination of the azimuth scale and level vial of the adapter mount as specified in 3.5.6.

4.6.2.7 Cable assembly.- Utilizing a 1/2 inch mandrel firmly secured in a holding medium, the approximate center of the cable length shall make initial contact on the top surface of the mandrel and equal pressure shall be applied at the cable extremes to effect the 270° bend. Subsequent to ten bends (minimum) there shall be no evidence of damage as a result of a visual and a continuity test to insure compliance with 3.5.7.

4.6.3 Electrical.

4.6.3.1 Continuity.- Standard measuring equipment shall be utilized to determine electrical continuity requirements of 3.6.1 of schematic Drawing F10540515.

4.6.3.2 Insulation resistance.- Measure the insulation resistance between mutually insulated parts by means of a megohm bridge or other approved method for conformance to 3.6.2.

4.6.4 Interface requirements.- Perform this test utilizing special testing equipment conforming to 4.5.2.3.1, two test housings F10540511 and two microscope bodies F10540942. One housing and body shall be designed to the maximum limit of each interface with a tolerance of 10% of total limit and one housing and body designed to the minimum limit of each interface with a tolerance of 10% of total limit. Do not extend any interface beyond maximum or minimum limits. All pertinent interfaces shall be provided as depicted on Drawings F10540511 and F10540942. The test housings and bodies shall be capable of being properly positioned and secured to the yoke assembly. The yoke assembly Connector C10540586 shall be engaged to a test connector with pertinent interfaces conforming to the requirements of connector Receptacle C10540587. Continuity between

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the connector and test connector receptacle shall be indicated when fully engaged. Continuity through the electrical contacts between the yoke assembly and adapter mount gage shall be maintained when the locking knob of the adapter mount is rotated with a torque value specified in 3.5.1 to its maximum clockwise excursion. When the locking knob is rotated to its maximum counterclockwise excursion, the yoke assembly shall be capable of being removed without restraint from the adapter mount.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing, and marking.- Packaging, packing, and marking shall be in accordance with Packaging Data Sheet 10540507. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use.- The yoke assembly secures and supports the range finder on the adapter mount and contains a microscope for viewing an azimuth scale.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Order of document precedence.
(c) Selection of applicable levels of preservation, packaging, and packing.
(d) Provisions for First Article testing.

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